

Point-by-Point Response to Reviewers

Protein-Conditional Normalizing Flows on Manifolds for Backbone Torsion Angle Modeling

Machine Learning with Applications

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General Response

We sincerely thank the Editor-in-Chief, Associate Editor, and Reviewers for their careful evaluation of our manuscript and for the constructive comments that helped us improve both its scientific positioning and empirical evaluation. In the revised manuscript, we have clarified the scope and novelty of the protein-conditional formulation, explicitly distinguished our contribution from the underlying manifold-flow constructions of Rezende et al. (2020), and made the limitation of identity-based protein embeddings for unseen proteins clear in the Abstract, Introduction, Method, and Conclusion. We have also strengthened the empirical evaluation by adding a protein-conditional Riemannian flow-matching baseline, paired protein-level statistical comparisons, and real-data density-based clustering analyses on the SCOP datasets. In addition, we clarified the role of the synthetic experiments, the numerical use of the grid in the Hellinger-distance calculation, and the reproducibility materials. All substantive revisions are indicated in colored text in the revised manuscript. Below, we reproduce the editorial and reviewer comments and provide our point-by-point responses together with the corresponding locations in the revised manuscript.

Section 1-a: Complete Comments from the Editor-in-Chief

Reviewer Comment

Comment 1: I am writing to inform you of the editorial decision regarding your paper submission MLWA-D-26-00991. to MLWA: Major Revision. If you wish to submit a revised version of your manuscript, please read this entire editorial decision letter carefully and address all requirements thoroughly to avoid delays in the review process.

Your revised submission must include a “Detailed Response to Reviewers” document containing the six required sections listed below. Each section must be clearly labeled and presented in the exact order specified. In addition, please upload the complete revision package—including the revised manuscript and all supporting files—through the EM system.

Failure to include any required section or to comply fully with these instructions may result in rejection of the revised submission.

The six required sections must appear in the following order:

- Section 1-a: The two complete comments from the Editor-in-Chief
- Section 1-b: The authors’ point-by-point responses to the Editor-in-Chief
- Section 2-a: The two complete comments from the Associate Editor
- Section 2-b: The authors’ point-by-point responses to the Associate Editor
- Section 3-a: The complete comments from the Reviewers
- Section 3-b: The authors’ point-by-point responses to the Reviewers (organized by individual reviewer)

As a verification step, please answer the following question in your response document:

Does your revised submission fully comply with the required structure outlined above? Yes

☐ No ☐

Please also note that a Major Revision decision does not constitute a guarantee of eventual acceptance, nor does it preclude the possibility of rejection in a subsequent review round. This decision reflects the reviewers’ assessment that the manuscript has potential but requires significant improvement before further consideration. Each resubmission will be evaluated independently based on its own merits, and neither the reviewers nor the journal are bound to a favorable outcome upon revision.

You are not required to incorporate all references suggested by the reviewers. If you identify any inappropriate or excessive citation requests, please feel free to contact me directly.

Reviewer Comment

Comment 2: Please confirm (Yes ☐ No ☐) that your submission complies with the following five requirements:

- **Author Information:** The EM system and the title page embedded in the full manuscript PDF must contain each author’s full name, complete institutional affiliation (including country), and valid institutional email address. The title page should not be submitted separately. All information must be accurate and consistent.
- **Institutional Emails:** Personal email addresses (e.g., Gmail, Yahoo, Hotmail, or Outlook) are not acceptable. The corresponding author is responsible for ensuring complete and accurate author information.

- **Authorship:** Any authorship changes must be fully justified in the point-by-point response and supported by written consent from all authors, including those added or removed. Unjustified changes may result in rejection.
- **References:** The manuscript must follow the MLWA reference style, based on APA referencing style as specified on the MLWA homepage.
- **Manuscript Format:** The manuscript must comply with the MLWA Guidelines on Manuscript Format, including the required single-column layout.

To submit you're a complete and compliant revised manuscript, please log in as an author at <https://www.editorialmanager.com/mlwa/>, and navigate to the "Submissions Needing Revision" folder.

To submit you're a complete and compliant revised manuscript, please log in as an author at <https://www.editorialmanager.com/mlwa/>, and navigate to the "Submissions Needing Revision" folder.

Section 1-b: Authors' Point-by-Point Responses to the Editor-in-Chief

Response

Response to Comment 1.

We thank the Editor-in-Chief for clearly specifying the requirements for the revised submission. We have followed the required structure for the "Detailed Response to Reviewers" document and organized it in the exact sequence requested: Section 1-a contains the two complete comments from the Editor-in-Chief; Section 1-b contains our point-by-point responses to the Editor-in-Chief; Section 2-a contains the two complete comments from the Associate Editor; Section 2-b contains our point-by-point responses to the Associate Editor; Section 3-a contains the complete comments from the Reviewers; and Section 3-b contains our point-by-point responses organized separately by reviewer.

We have also revised the manuscript in response to the editorial and reviewer comments and will submit the complete revision package, including the revised manuscript and supporting files, through the Editorial Manager system.

Verification: Does your revised submission fully comply with the required structure outlined above? **Yes** ☒ **No** ☐

Response

Response to Comment 2. **Yes** ☒ **No** ☐

We confirm that the revised submission complies with the five requirements specified by the Editor-in-Chief. The full names, complete institutional affiliations including country, and valid institutional e-mail addresses of all authors have been carefully verified for consistency between the manuscript and the Editorial Manager system. All authors use institutional e-mail addresses.

No changes have been made to the authorship of the manuscript. Specifically, the number of authors, author order, and institutional affiliations remain unchanged from the original submission; therefore, no authorship-change justification or consent documentation is required.

The manuscript is prepared in the required single-column format. The reference list has been carefully checked and formatted to conform to the MLWA APA-based reference style, including alphabetical organization and verification of bibliographic information and DOI information where available.

Location: Title page and author information; reference list; manuscript-wide formatting.

Section 2-a: Complete Comments from the Associate Editor

Reviewer Comment

Comment 1: Thank you for submitting your manuscript to MLWA. Following a careful review, the reviewers have identified several substantive issues that must be adequately addressed before the manuscript can be considered further. Please revise the manuscript thoroughly and provide a clear, point-by-point response to each reviewer comment. All revisions should be clearly indicated using colored text in the manuscript. Kindly ensure that the track changes function is turned off prior to submission.

Reviewer Comment

Comment 2: Please confirm (Yes ☐ No ☐) whether Section #1-a of your revised submission includes the complete and unaltered text of two comments from the Editor-in-Chief. Please note that the complete and verbatim inclusion of the Editor-in-Chief's comments is a mandatory journal requirement. Any omission or modification of these comments is considered a breach of editorial ethics and is not acceptable.

Section 2-b: Authors' Point-by-Point Responses to the Associate Editor

Response

Response to Comment 1.

We thank the Associate Editor for the careful assessment of our manuscript and for emphasizing the importance of a thorough revision. We have carefully considered all comments raised by the reviewers and have undertaken a substantial revision of the manuscript accordingly. In particular, we have prepared detailed, point-by-point responses to each reviewer comment in Section 3-b of this document. The corresponding changes in the revised manuscript are clearly indicated using colored text so that they can be readily identified by the Editor and Reviewers. The final revised manuscript will be submitted with the track-changes function turned off, as requested.

The substantive scientific issues raised by the reviewers, including the scope and novelty of the protein-conditional formulation, its limitation with respect to unseen proteins, comparison with a modern generative-modeling baseline, statistical assessment of model performance, real-data clustering validation, and reproducibility, are addressed individually in the responses to the reviewers and through corresponding revisions to the manuscript.

Location: The revisions are indicated in colored text throughout the revised manuscript, and detailed locations of individual changes are reported under the corresponding responses in Section 3-b.

Response

Response to Comment 2. Yes ☒ No ☐

We confirm that Section 1-a of this response document includes the complete and unaltered text of both comments from the Editor-in-Chief. The comments have been reproduced verbatim from the editorial decision letter, without omission, summarization, paraphrasing, or substantive modification.

We have retained the complete wording of the Editor-in-Chief’s comments, including the submission instructions and compliance requirements, to ensure full adherence to the journal’s mandatory response-document structure.

Location: Section 1-a, “Complete Comments from the Editor-in-Chief,” Comments 1 and 2.

Section 3-a: Complete Comments from the Reviewers

Reviewer 1

Reviewer Comment

Comment 1: The core idea regarding conditioning normalizing flows on learned embeddings is conceptually straightforward. The heavy theoretical lifting (manifold flows on circles, tori, spheres) is largely borrowed from Rezende et al. [2020]. The protein-conditional extension is useful but incremental.

Reviewer Comment

Comment 2: The embedding-based conditioning mechanism (a simple lookup table) is lightweight to the point of being limiting. The authors explicitly acknowledge this cannot generalize to unseen proteins—a major practical constraint that undermines real-world applicability.

Reviewer Comment

Comment 3: In order to strengthen the empirical foundations, the authors should:

1. Addition of at least one modern baseline (e.g., flow matching or score-based manifold model).

Reviewer Comment

Comment 4: In order to strengthen the empirical foundations, the authors should:

2. Explicit discussion of the zero-shot limitation in the abstract/conclusion, not just Section 2.5.

Reviewer Comment

Comment 5: In order to strengthen the empirical foundations, the authors should:

3. Real-data clustering validation on SCOP datasets (currently only synthetic).

Reviewer Comment

Comment 6: Considering the scale between 1,591 proteins and $\sim 732\text{K}$ residues (SCOP Easy) is modest by contemporary standards. AlphaFold2-era structural biology operates at hundreds of millions of structures.

This work is unlikely to shift practice in protein structure modeling. It serves more as a principled statistical foundation than a tool practitioners will immediately adopt.

Reviewer 2

Reviewer Comment

Comment 1: The manuscript presents a technically sound extension of neural circular spline flows to protein-conditional settings. Still, its novelty is severely limited because it directly adopts the core manifold flow architecture from Rezende et al. (2020) with only minor modifications (learned protein embeddings).

Reviewer Comment

Comment 2: The GitHub repository link is non-functional (CFFF), raising reproducibility concerns.

Reviewer Comment

Comment 3: The conditioning mechanism via protein embeddings fails to generalise to unseen proteins, contradicting the claimed promise of "protein-specific" modelling.

Reviewer Comment

Comment 4: Additionally, the synthetic experiments use overly simplistic distributions (wrapped normal mixtures, conditional von Mises) that poorly represent the complex multimodal nature of real protein torsion angles.

Reviewer Comment

Comment 5: At the same time, the real-data improvements (NLL differences of 0.05-0.1) are modest and not statistically validated.

Reviewer Comment

Comment 6: The clustering application, while interesting, uses density estimates on fixed grids rather than the exact likelihood formulation the paper claims as its advantage.

Section 3-b: Authors’ Point-by-Point Responses to the Reviewers

Reviewer 1

Reviewer Comment

Comment 1: The core idea regarding conditioning normalizing flows on learned embeddings is conceptually straightforward. The heavy theoretical lifting (manifold flows on circles, tori, spheres) is largely borrowed from Rezende et al. [2020]. The protein-conditional extension is useful but incremental.

Response

We thank the Reviewer for this assessment and agree that the underlying manifold normalizing-flow machinery is not a novel contribution of the present work. Specifically, the circular spline transformations on $\mathbb{S}^1$, their extension to product manifolds such as $\mathbb{T}^D$, and the sphere–cylinder construction on $\mathbb{S}^D$ build on the manifold-flow framework of Rezende et al. (2020). We have revised the manuscript to make this boundary of novelty explicit and to avoid implying that the underlying geometric construction itself is introduced here.

The methodological contribution of the present study is instead the *protein-conditional* formulation built on this framework, in which learnable protein-specific embeddings are jointly optimized with the flow parameters. This allows a single shared manifold-aware model to represent distinct protein-specific torsion-angle distributions while retaining tractable likelihood evaluation.

We have also strengthened the empirical positioning of this contribution. In response to the Reviewer’s recommendation, we added a direct comparison with a modern protein-conditional Riemannian flow-matching baseline under a matched experimental protocol. This additional experiment allows the revised manuscript to position PC-NCSF empirically relative to a more recent continuous-time manifold generative-modeling paradigm rather than relying solely on comparisons with the unconditional NCSF baseline.

Location: Introduction, immediately following the list of main contributions; Section 3.1.3, “Comparison with a Modern Flow-Matching Baseline,” and Table 3.

Reviewer Comment

Comment 2: The embedding-based conditioning mechanism (a simple lookup table) is lightweight to the point of being limiting. The authors explicitly acknowledge this cannot generalize to unseen proteins—a major practical constraint that undermines real-world applicability.

Response

We agree with the Reviewer that the current protein-identity embedding imposes an important zero-shot limitation. Because the conditioning representation is a learned lookup embedding associated with proteins observed during training, the present model cannot directly generate a protein-specific density for a previously unseen protein without additional fitting. We therefore agree that this limitation should not be confined to the methodological description.

In the revised manuscript, we have made this limitation explicit in both the Abstract and the Conclusion and have moderated the discussion of practical generalizability accordingly. We now distinguish more clearly between protein-specific density estimation for proteins

represented during model training and zero-shot prediction for unseen proteins, which is not supported by the current architecture.

We also identify replacing the identity-based lookup embedding with a conditioning representation derived from transferable protein information, such as sequence- or structure-derived features, as an important direction for future work.

Location: Abstract; Section 2.5, “Conditional Modeling via Learned Protein Embeddings”; and Section 4, “Conclusion” (Limitations).

Reviewer Comment

Comment 3: In order to strengthen the empirical foundations, the authors should:

1. Addition of at least one modern baseline (e.g., flow matching or score-based manifold model).

Response

We agree and have added a modern flow-matching baseline to the revised manuscript. Specifically, we implemented a Protein-Conditional Riemannian Flow Matching (PC-RFM) model and evaluated it under the same protein-conditioning setting and the same train/validation split used for the corresponding PC-NCSF experiment.

On the paired protein-level comparison, 493 proteins were represented in the validation outputs of both methods. PC-NCSF achieved a lower mean validation NLL than PC-RFM for 443 of these 493 proteins (89.9%). The paired difference was statistically significant using both the Wilcoxon signed-rank test ($p = 3.8 \times 10^{-69}$) and the paired t -test ($p = 1.1 \times 10^{-85}$), with a large standardized paired effect (Cohen’s $d = 1.09$).

Importantly, the comparison is performed at the *protein level* rather than treating individual residues as independent observations. This provides a more appropriate assessment of the consistency of the performance difference across proteins and directly addresses the need for statistical validation of the model comparison.

These results have been added as a new experiment and are reported in the revised manuscript.

Location: Section 3.1.3, “Comparison with a Modern Flow-Matching Baseline,” and Table 3.

Reviewer Comment

Comment 4: In order to strengthen the empirical foundations, the authors should:

2. Explicit discussion of the zero-shot limitation in the abstract/conclusion, not just Section 2.5.

Response

We agree. As also noted in our response to Comment 2, the inability of the current lookup-based conditioning mechanism to generalize directly to unseen proteins is an important limitation of the proposed formulation.

We have therefore added an explicit statement of this limitation to both the Abstract and the Conclusion. These revisions make clear that the current protein-conditional formulation provides protein-specific densities for proteins represented during training but does not constitute a zero-shot model for previously unseen proteins.

Location: Abstract; Section 4, “Conclusion” (Limitations).

Reviewer Comment

Comment 5: In order to strengthen the empirical foundations, the authors should:
3. Real-data clustering validation on SCOP datasets (currently only synthetic).

Response

We agree that evaluating the density-based clustering procedure on real structural data provides a substantially more informative assessment of its downstream utility. We have therefore extended the Hellinger-distance and Ward-linkage clustering pipeline, previously demonstrated only in the controlled synthetic experiments, to all four real SCOP benchmark tiers.

Unlike the near-perfect separation observed in some controlled synthetic settings, clustering performance on the real SCOP datasets was weak, with Adjusted Rand Index (ARI) values ranging from 0.001 to 0.097 across the four tiers. We additionally examined alternative linkage choices and found that the overall conclusion was unchanged.

We have deliberately retained and reported these results despite their weak performance because they provide an important qualification of the original downstream-utility claim. The revised manuscript now makes clear that improved protein-specific density estimation does not necessarily imply that the resulting density functions form strongly cluster-separable representations of complex real protein structural classes.

Accordingly, we have moderated the interpretation of the clustering experiments. Accurate manifold-aware density estimation is now presented as the primary contribution of the work, whereas density-based clustering is treated as a secondary exploratory application. The strong synthetic clustering results remain useful as a controlled demonstration that the estimated densities can recover deliberately separated distributional structure, but they are no longer presented as evidence of comparable classification performance on real SCOP data.

Location: Section 3.5, “Real-Data Clustering Validation on SCOP,” Table 8; and Section 4, “Conclusion” (Limitations).

Reviewer Comment

Comment 6: Considering the scale between 1,591 proteins and $\sim 732\text{K}$ residues (SCOP Easy) is modest by contemporary standards. AlphaFold2-era structural biology operates at hundreds of millions of structures.

This work is unlikely to shift practice in protein structure modeling. It serves more as a principled statistical foundation than a tool practitioners will immediately adopt.

Response

We appreciate this perspective and agree that the scale of the datasets considered in this study is modest relative to contemporary large-scale protein-structure resources. The objective of the present work, however, is not large-scale protein structure prediction or foundation-model training. Rather, it is to investigate accurate likelihood-based density estimation for manifold-valued protein torsion-angle distributions and to determine whether protein conditioning improves such density estimates within a shared geometrically appro-

priate model.

We have revised the manuscript to make this intended scope more explicit and to avoid overstating the immediate practical impact of the method. In particular, the Conclusion now characterizes the proposed framework as a statistical modeling approach for protein-specific angular distributions, while recognizing that evaluation on substantially larger structural collections and the development of transferable conditioning mechanisms will be necessary before broader practical applicability can be established.

Location: Section 4, “Conclusion” (Limitations).

Reviewer 2

Reviewer Comment

Comment 1: The manuscript presents a technically sound extension of neural circular spline flows to protein-conditional settings. Still, its novelty is severely limited because it directly adopts the core manifold flow architecture from Rezende et al. (2020) with only minor modifications (learned protein embeddings).

Response

We thank the Reviewer for this important comment. We agree that the underlying manifold normalizing-flow constructions are not a novel contribution of the present work. In particular, the circular, toroidal, and spherical geometric constructions used by PC-NCSF build on the framework of Rezende et al. (2020). We have therefore revised the Introduction to make this boundary of novelty explicit and to avoid attributing the underlying manifold-flow architecture itself to the present study.

Our contribution is more specifically the protein-conditional formulation: learnable protein-identity embeddings are jointly optimized with a shared manifold flow, allowing a single model to yield distinct, individually queryable protein-specific angular density estimates while sharing statistical information across proteins. We have revised the manuscript to state this contribution more precisely and have moderated the corresponding novelty claims. To strengthen the empirical assessment of whether this conditional formulation provides value beyond the underlying manifold-flow construction, we have also added a new comparison with Protein-Conditional Riemannian Flow Matching (PC-RFM), a more recent continuous-time manifold generative-modeling approach. Under the matched experimental setting, PC-NCSF achieved lower mean validation NLL for 443 of the 493 proteins (89.9%), with the paired difference supported by both the Wilcoxon signed-rank test ($p = 3.8 \times 10^{-69}$) and the paired t -test ($p = 1.1 \times 10^{-85}$), and a large paired effect size (Cohen’s $d = 1.09$). These results are now reported in the new Section 3.1.3.

Accordingly, we do not position PC-NCSF as a new manifold-flow geometry, but rather as a protein-conditional adaptation and empirical evaluation of established manifold-flow machinery for protein-specific angular density estimation.

Location: Introduction, immediately following the list of main contributions; Section 3.1.3.

Reviewer Comment

Comment 2: The GitHub repository link is non-functional (CFFF), raising reproducibility concerns.

Response

We thank the Reviewer for identifying this issue. We have corrected the repository accessibility problem and verified that the repository is now publicly accessible at <https://github.com/mamintoosi-papers-codes/CFFF>.

We have also reviewed the repository for consistency with the revised manuscript. In particular, an internal naming inconsistency in which some training scripts referred to the conditional model as “PC-FFF” has been corrected to “PC-NCSF”, and the repository documentation and citation metadata have been updated accordingly. These revisions are intended to make the computational materials easier to identify and use alongside the manuscript and to improve the reproducibility of the reported experiments.

Location: Abstract (implementation and code availability statement); GitHub repository and repository documentation.

Reviewer Comment

Comment 3: The conditioning mechanism via protein embeddings fails to generalise to unseen proteins, contradicting the claimed promise of “protein-specific” modelling.

Response

We thank the Reviewer for highlighting this important distinction. We agree that the current protein-identity embedding is a learned lookup representation and therefore does not support zero-shot generalization to proteins that were absent during model training. We have revised the manuscript to state this limitation explicitly and to clarify the intended scope of the term “protein-specific.”

In the present formulation, “protein-specific” refers to the model’s ability to learn and provide individually queryable conditional density estimates for each protein represented during training, while sharing the parameters of the underlying manifold flow across proteins. It is not intended to imply that PC-NCSF can infer the conformational density of a previously unseen protein from its sequence or structure.

To prevent this distinction from being overstated, we now explicitly identify the lack of zero-shot generalization as a limitation in the Abstract, Introduction, and Conclusion. We also identify replacing the identity-based lookup embedding with sequence- or structure-derived conditioning representations as an important direction for future work.

Location: Abstract; Introduction, immediately following the clarification of the contribution and novelty; Section 2.5; and Conclusion (Limitations).

Reviewer Comment

Comment 4: Additionally, the synthetic experiments use overly simplistic distributions (wrapped normal mixtures, conditional von Mises) that poorly represent the complex multimodal nature of real protein torsion angles.

Response

We thank the Reviewer for this important comment. We agree that controlled synthetic distributions cannot reproduce every aspect of the heterogeneity and multimodal complexity observed in empirical protein torsion-angle data. However, we would like to clarify that the

synthetic settings were not chosen as arbitrary simplified distributions. In particular, the first simulation setting is motivated by the characteristic bimodal angular structure of consecutive C_α atoms reported by Hamelryck et al. (2006), as already described in Section 3.2.1.

The purpose of the simulation study is to provide protein-relevant angular settings in which the generating densities and group structure are known. This makes it possible to directly evaluate recovery of the protein-specific densities and to determine whether density-based clustering recovers the known ground-truth groups—assessments that cannot be performed in the same way on empirical protein data because the underlying densities are unknown.

To make this rationale clearer, we have revised the opening paragraph of Section 3.2 to distinguish the role of the synthetic experiments from that of the real-data analyses. The synthetic experiments are intended as controlled methodological evaluations that complement, rather than replace, the experiments on real protein data in Section 3.3 and the real-data density-based clustering analysis in Section 3.5.

Location: Section 3.2 (opening paragraph).

Reviewer Comment

Comment 5: At the same time, the real-data improvements (NLL differences of 0.05-0.1) are modest and not statistically validated.

Response

We thank the Reviewer for this important observation. We agree that the magnitude of the NLL improvement should be distinguished from its statistical support. To address this concern, we performed a paired protein-level analysis comparing the unconditional model with the conditional model using $d_{\text{emb}} = 16$.

Specifically, we computed validation NLL separately for each protein and performed paired comparisons across the 493 proteins represented in both validation outputs. The conditional model achieved lower NLL for 283 of 493 proteins (57.4%), with a mean paired improvement (unconditional minus conditional NLL) of 0.060 nats. The paired difference was statistically significant under both the Wilcoxon signed-rank test ($p = 5.3 \times 10^{-9}$) and the paired t -test ($p = 2.3 \times 10^{-12}$), with a small standardized paired effect (Cohen’s $d = 0.32$).

We have added these results to Section 3.1.2. We have also revised the interpretation to distinguish statistical evidence for a systematic improvement from its modest effect magnitude, rather than relying solely on the aggregate validation NLL.

Location: Section 3.1.2, immediately following the comparison of embedding dimensions with the unconditional baseline.

Reviewer Comment

Comment 6: The clustering application, while interesting, uses density estimates on fixed grids rather than the exact likelihood formulation the paper claims as its advantage.

Response

We thank the Reviewer for raising this point. We agree that the original description could give the impression that the learned densities themselves were approximated on a fixed grid. This is not the case.

The fixed grid is used only to numerically approximate the Hellinger-distance integral required for the downstream clustering analysis. At each grid point, the protein-specific density is evaluated directly from the trained PC-NCSF model using its invertible change-of-variables formulation; the grid is not used to estimate or approximate the underlying density function. Thus, the numerical approximation concerns the integration required to obtain pairwise Hellinger distances, rather than the pointwise density or likelihood evaluation provided by PC-NCSF.

We have revised Section 3.2.3 to make this distinction explicit and to remove the potentially ambiguous wording that the estimated densities are “evaluated on a uniform grid.”

Location: Section 3.2.3, Density-Based Clustering (Clustering procedure).